

Semiconductor

Questions with Fully Worked Answers · FormulaVerse PYQ Bank

Q1. In the given circuit, a diode D is connected to an external resistance $R = 100\ \Omega$ and an e.m.f. of 3.5 V. If the barrier potential developed across the diode is 0.5 V, the current in the circuit will be ____ mA.

JEE Main (Physics, Semiconductor Devices)

✓ **Answer: 30**

Solution:

Once the diode is forward biased and conducting, the source e.m.f. is shared between the barrier potential and the resistor:

$$I = (V - V_{\text{barrier}})/R = (3.5 - 0.5)/100 = 3/100 = 0.03\text{ A} = 30\text{ mA}.$$

Q2. A Zener diode with 5 V zener voltage is used to regulate an unregulated dc voltage input of 25 V. For a $400\ \Omega$ resistor connected in series, the zener current is found to be 4 times the load current. The load current (I_L) and load resistance (R_L) are ____.

JEE Main (Physics, Semiconductor Devices)

✓ **Answer: $I_L = 10\text{ mA}$, $R_L = 500\ \Omega$**

Solution:

$$\text{Voltage across the series resistor} = V_{\text{in}} - V_Z = 25 - 5 = 20\text{ V}.$$

$$\text{Total current through the series resistor: } I = 20/400 = 0.05\text{ A} = 50\text{ mA}.$$

$$\text{This splits between the zener and the load: } I = I_Z + I_L = 4I_L + I_L = 5I_L.$$

$$\text{So } I_L = 50/5 = 10\text{ mA}.$$

$$R_L = V_Z/I_L = 5/0.01 = 500\ \Omega.$$

Source: JEE Main 2026 (NTA official releases, physics section) · formulaverse.in